

Chenyan Jing

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Portfolio: <https://jingchenyan3030.github.io/>

Github: <https://github.com/jingchenyan3030>

Education:

Johns Hopkins University, Baltimore, MD, USA

M.S.E. in Robotics

May 2026

Shanghai Maritime University, Sino-Dutch Mechatronics Engineering College, Shanghai, China

B.Eng in Mechanical and Electronic Engineering

June 2024

Technical skills:

Languages: Python, C++, MATLAB

Computer Vision: Foundation Models, Monocular Depth Estimation, 3D Back-Projection, Camera Calibration

Deep Learning: PyTorch, UNet, ViT, Transformer, SAM3, YOLOv8-Pose, HRNet, Depth Anything V2

Engineering Tools: OpenCV, NumPy, Git, Linux, ROS2, Multi-GPU Training, Model Inference, Evaluation

Robotics: Robot Kinematics, Resolved-Rate Control, Sensor Integration, Arduino

Experience:

Research Assistant, ARCADE Lab, Johns Hopkins University

Dense Structural Priors for Sparse Functional Landmark Localization in Surgical Videos

- Developed a multi-frame ViT-based heatmap localization framework for action-dependent surgical tool-tip and anchor prediction across clipping, grasping, and dissection.
- Leveraged coarse landmark predictions to point-prompt zero-shot SAM 3 masks, then fused visual features, coarse heatmaps, and structural mask priors through a lightweight refinement adapter.
- Implemented an auxiliary mask-supervision branch to inject dense instrument geometry.
- Evaluated on 7,867 clips from 60 surgical videos across five datasets; improved overall tip F1 from 52.9% to **72.4%** and anchor F1 from 51.5% to **58.0%**.

AffordTissue: Tool-Action Conditioned Tissue Affordance Prediction

- Contributed to a multimodal framework for predicting safe tissue-interaction regions for a given surgical tool-action pair from temporal laparoscopic videos.
- Encoded temporal video features with Video Swin and structured surgery-tool-action prompts with SigLIP.
- Used AdaLN-based conditional modulation to fuse language and visual features for affordance heatmap prediction; conducted ablations on language, temporal context, and architecture design.
- Achieved **20.6 px ASSD** and **0.667 PCK@0.1**, outperforming Molmo-VLM (60.2 px ASSD) and Qwen-VLM (111.3 px ASSD) for dense tissue-affordance localization.

Predictive Retinal Motion Compensation for Robotic Subretinal Injection

- Used U-Net ILM/RPE axial depth trajectories from iOCT as temporal inputs for next-step retinal motion prediction.
 - Implemented and benchmarked LSTM and Mamba models using 20-frame histories of retinal depth and inter-frame timing for next-frame ILM/RPE prediction; Mamba achieved 33.52 μm RMSE / 25.44 μm MAE, outperforming LSTM at 42.61 μm RMSE / 32.95 μm MAE.
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Selected Projects

3D Vertebrae Segmentation Post-Processing

- Developed an anatomy-aware 3D post-processing pipeline to refine fragmented and anatomically inconsistent vertebrae segmentations through connected-component analysis and component-level keypoint localization.
- Fit a robust spinal centerline using RANSAC over component keypoints, then used centerline-relative position and axial spatial constraints to estimate vertebral ordering and identify plausible neighboring components.
- Constructed a weighted component graph and applied BFS-based traversal to reconnect fragmented vertebral regions, with penalty terms for implausible spatial links, label jumps, and duplicate assignments.
- Improved average Dice similarity coefficient from 77.5% to 93% on vertebral segmentation data.

Language-Conditioned Visual Action Policy for Robotic Manipulation | In Progress

- Developing a minimal vision-language-action policy in ManiSkill that fuses RGB observations, robot proprioception, and natural-language instructions to generate action chunks via flow matching.
- Training and evaluating on PickCube, PushCube, and StackCube using paraphrased language instructions and official ManiSkill demonstrations.
- Benchmarking language-conditioned flow-matching policies against no-language and direct-regression baselines using rollout success rate, action prediction error, and qualitative multi-task rollouts.

Estimating 3D Human Pose Trajectories from Monocular Video

- Built a monocular 3D human-pose reconstruction pipeline by combining YOLOv8-Pose 2D keypoints, Depth Anything relative depth estimation, camera calibration, and geometric back-projection with scale recovery.

- Converted frame-wise detections into temporally ordered 3D pose trajectories; evaluated against RGB+LiDAR references and analyzed failure modes including depth ambiguity, scale drift, occlusion, and calibration mismatch.

LiDAR-Based Particle Filter Localization with Custom AMCL Models

- Implemented custom odometry and laser-beam sensor models for a Nav2 AMCL particle-filter localization pipeline in a simulated indoor environment.
 - Developed a noisy odometry motion model that decomposed robot motion into rotation–translation–rotation components and sampled Gaussian motion uncertainty using tunable error parameters.
 - Implemented a probabilistic LiDAR beam model combining normalized hit, unexpected-short, max-range, and random-return distributions, including robust handling of NaN, Inf, and out-of-range measurements.
 - Transformed LiDAR extrinsics from the robot frame to the map frame and used occupancy-map ray tracing to compute expected ranges for each particle hypothesis.
 - Updated particle likelihoods using scan-level beam probabilities and integrated the models into the AMCL sensor-update and resampling workflow for pose convergence in RViz.
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Internship

R&D Assistant Intern, Newstart Robot System Technology Co., Ltd., Shandong, China

- Assisted with mechanical design and feasibility analysis of soft robotic gripper systems using SolidWorks, supporting prototype development and design iteration.
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Publications

- Chenyan Jing, et al. Dense Structural Priors for Sparse Functional Landmark Localization in Surgical Videos. Under review in MICCAI Workshop Medagi.
- Co-author of AffordTissue: Dense affordance prediction for surgical scene understanding. Accepted at MICCAI.
- Co-author of SAW: Surgical action world modeling for surgical video understanding. Accepted at MICCAI.
- Co-author of Predictive retinal motion compensation for robotic subretinal injection. Under review in IEEE.